

HCI 594 Capstone
Winter 2025

Final Project Report
Project: KidsConnect
Team: TAS Group 1

Table of Contents

Team/Project Information	3
Abstract	3
Introduction	3
Goals	4
Goal 1: Validate the usability of the parent interface.	4
Goal 2: Evaluate the effectiveness of the reward system in meeting parents' expectations.	4
Methods	4
Method 1: Surveys	4
Method 2: Interviews	5
Method 3: Usability Testing	5
Results	6
Method 1: Surveys	6
Method 2: Interview	7
Method 3: Usability Testing	8
Discussion	8
Evaluation of Project Goals	8
Implications of Design	9
What Could Have Done Differently	9
Opportunities for Future Work	9
Conclusion	9
References	9
Appendix 1: Supporting Material	11
Survey Theme	11
Interview Coding	12
Affinity Diagram	13
User Personas	14
Usability Testing Round 1 Result	15
ASQ questionnaire Round 1	15
Usability Testing Round 2 Result	17
ASQ questionnaire Round 2	17
Heat Map for adding/deleting child profile Scenario	19
Site Map or Flow Diagram	20
Interactive Prototype	25
Prototype Links/Instructions	25
User Interview Template/Script	29
Interview Protocol (Teacher)	29
Interview Protocol (Couple)	30
Interview Protocol (Parent)	31

Team/Project Information

Team Name: TAS (Group 1)

Project Name: KidsConnect: Simplifying Family Task Management

Team Members:

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Abstract

KidsConnect is a mobile chore management application designed for parents, teachers, caregivers, and married couples to simplify the process of assigning and tracking tasks with the addition of a customizable gamified reward system. The project included surveys, interviews, literature reviews, and usability testing to understand user needs and adjust overall designs. Through surveys (n=23), interviews (n=6), and two rounds of usability testing (n=12) the project validated key design decisions and identified critical refinements for the app's main features of assigning tasks, creating rewards, adding and deleting children profiles, and sending task reminders. Meanwhile, in surveys, 81% of users desire a customizable reward system that is nonmonetary, and 68.8% want automated chores based on age. Interviews highlighted the use of informal methods like verbal agreements and physical task lists with the desired features such as synced and shared calendars and voice command integration.

Usability testing conducted in two rounds provided critical design issues with profile navigation, specifically a confusion between personal and kid specific profiles while some revisions led to great improvements in navigation, adding and deleting child profiles (ASQ scores improved from 4.10 to 5.76 and 6.30). However, a new challenge with the task assignment feature (ASQ score of 3.70), prompted recommendations for adjusting the onboarding process with guided tutorials to first time users.

The findings underscore KidsConnect's success in addressing diverse user needs through inclusive design while highlighting opportunities for AI driven task recommendations and real time collaboration. Future iterations will focus on onboarding tutorials and testing with children to further understand and adjust engagement strategies and align some features for children's needs. This research raises the potential of gamified, family centered tools to increase responsibility and collaboration in children's daily routines.

Introduction

KidsConnect is a mobile task management application designed to help parents, teachers, and caregivers to organize household chores and tasks for children easily and more efficiently. The project idea was developed to provide a quick, simple, and effective solution for assigning, tracking, and rewarding task completion for children. Research, prototyping, and usability testing were conducted to ensure the app's features aligns with users needs and to make any refinement based on feedback. Unlike existing applications from the competitive analysis that was done. Other apps focused primarily on financial rewards, KidsConnect differentiates itself by prioritizing a friendly, gamified experience with no financial incentives, customizable chore tracking, and a minimalistic user interface. The competitive analysis showed that while other apps offer chore management and financial education, they lack engagement with gamification and family collaboration features. KidsConnect stands out by including gamified elements such as points, levels, and badges, as well as custom rewards system.

Literature review findings support this approach, showing that gamification improves children's motivation (Sarour et al., 2023), mastery-based rewards enhance responsibility (Madjar et al., 2016), and parental involvement strengthens behavioral and academic development (Roy & Giraldo-García, 2018). Focusing on the principles of user centered design, and accessibility, KidsConnect's idea is to provide an easy, quick, and effective task management system.

Goals

Goal 1: Validate the usability of the parent interface.

- Measure: Conduct usability testing with 12 participants to evaluate the ease of assigning chores, setting rewards, and tracking progress. Collect feedback on clarity, navigation, and overall satisfaction.
- Revisions or adjustments: Expanded usability testing participant demographics to include couples and guardians.
- Explain changes: The expansion ensures inclusivity and accommodates varying task management needs.

Goal 2: Evaluate the effectiveness of the reward system in meeting parents' expectations.

- Measure: Conducted surveys(23 participants) and interviews(6 participants) to gather feedback on whether the proposed reward system aligns with parenting styles.
- Revisions or adjustments: Adjusted survey questions to reflect varied household and group dynamics.
- Explain changes: A broader user base required refining the reward system to cater to different relational dynamics and task delegation strategies.

Methods

Method 1: Surveys

Goal (contribution to project): Gather data on user preferences and task management challenges.

Revisions or adjustments: Expanded participation pool from parents to include teachers, couples, and guardians.

Explain changes: The survey was modified to address the different needs of various user groups.

- The survey was distributed to parents, teachers, couples, and guardians who manage household or classroom tasks.
- Participants were recruited through social media groups, community forums, and personal networks.
- A total of 16 respondents completed the survey, representing a mix of parents with young children, teachers managing classroom chores, and couples sharing household responsibilities.

Data Collection Methods

- The survey was conducted using Google Forms and distributed online.
- Participants answered multiple-choice questions, Likert-scale ratings, and open-ended questions about their current chore management practices, challenges, and preferences.
- The survey took approximately 5-10 minutes to complete.

Data Analysis Methods

- Quantitative data from multiple-choice and Likert-scale questions were analyzed using descriptive statistics, such as response frequencies and percentages.
- Qualitative responses (from open-ended questions) were reviewed and categorized into key themes (see [Survey Theme](#)).

Tools Used

- Google Forms
- Figjam

Method 2: Interviews

Goal (contribution to project): Validate pain points and feature requirements.

Revisions or adjustments: Added interview questions addressing different task management scenarios for various relationship dynamics.

Explain changes: The interview questions were refined to gather insights into how different user groups manage and assign tasks.

- Participants were recruited through personal networks and professional communities, including parents, teachers, couples, and guardians responsible for chore management.
- A total of six participants were interviewed, each representing different household structures and responsibilities, including teachers managing classroom chores, couples sharing household tasks, and parents organizing family responsibilities. ([User Interview Template/Script](#))

Data Collection Methods

- The interviews were conducted remotely via Zoom to allow for flexible scheduling and participation.
- Each session lasted approximately 30–45 minutes, covering topics such as chore assignment methods, accountability systems, motivational strategies, and preferences for digital solutions.
- The interview format was semi-structured, allowing for open-ended discussions while maintaining consistency across participants.
- Sessions were recorded (with participant consent) for accuracy, and key insights were transcribed for analysis.

Data Analysis Methods

- Thematic analysis was conducted by categorizing responses into major themes, such as chore management practices, challenges, rewards, and technology adoption.
- Affinity diagramming was used to group participant responses and identify shared patterns across different demographics (see Appendix: [Affinity Diagram](#)).
- Key insights were synthesized to highlight common pain points and feature requests relevant to digital chore management solutions (see Appendix: [Interview Coding](#)).

Tools Used

- Zoom: For conducting and recording the interviews.
- Google Docs: For transcribing
- Figma: Create An Affinity Diagram

Method 3: Usability Testing

Goal (contribution to project): Evaluate the interface and interactions for diverse user types.

Revisions or adjustments: None

Explain changes: None

- Participants were recruited from our target user groups, including parents, teachers, couples, and guardians who actively manage household or classroom tasks.
- A total of 12 users participated in the usability testing round one and two.
- The testing group was selected to ensure a diverse range of users with different experiences in chore management, digital tools, and task-tracking methods.

Data Collection Methods

- Two rounds of usability testing were conducted remotely using Maze to evaluate and refine the prototype.
- Round 1 (Class Activity - Peer Review): This initial round of usability testing was conducted as part of a class activity, where classmates evaluated the prototype. This round aimed to identify major usability

issues, test the core interactions, and gather initial feedback on the overall navigation and user experience before testing with actual target users.

- Round 2 (Target Users - Real-World Testing): After refining the prototype based on feedback from Round 1, the second round was conducted with actual target users (Married couples, and teachers) to ensure the design effectively met their needs. Participants were asked to complete key tasks, such as assigning chores, adding a custom reward, and managing children's profiles while screen-recording their sessions with audio to verbalize their thoughts.
- Each usability testing session lasted approximately 5-10 minutes, followed by an After Scenario Questionnaire (ASQ) to gather additional qualitative feedback.

Data Analysis Methods

- Round 1 Analysis (Class Activity): Focused on identifying usability pain points, navigation difficulties, and general interface issues based on peer feedback. Findings from this round guided early design refinements to improve clarity and flow before testing with real users.
- Round 2 Analysis (Target User): Measured task success rates, interaction patterns, and user satisfaction to assess the effectiveness of design improvements. This round helped validate whether the prototype addressed real-world user needs and highlighted any remaining areas for enhancement.
- Findings from both rounds were synthesized into usability challenges, iterative design changes, and final recommendations for optimizing the user experience.

Tool Used

- Maze: For conducting online usability testing and collecting analytics.

Results

Method 1: Surveys

The survey results offer insights into how different users manage household chores, assign tasks, track progress, and implement reward systems to encourage children or dependents.

Current Methods

Survey responses revealed that most users rely on verbal agreements to delegate tasks, with 56.3% using this method. A smaller percentage (12.5%) use written lists or digital tools, while 31.3% do not assign specific chores at all. While informal task management is common, these findings suggest there is interest in structured tools that could enhance organization and accountability.

Challenges

Users reported mixed experiences with tracking household chores. Some had no challenges, either because they did not assign tasks or their children completed chores as expected. However, those who faced difficulties mentioned issues such as forgetting to follow up on tasks, maintaining motivation, and fairly distributing workload between partners. Additionally, some noted that children struggle with accountability, highlighting the need for better tracking and engagement strategies.

Rewards

Survey responses showed that screen time and extra playtime were the most commonly used incentives, with 43.8% of respondents selecting each. Points, badges, and personalized rewards were also popular, while monetary rewards were used less frequently (18.8%). A small portion (6.3%) reported not using any reward system at all. These findings suggest that non-monetary rewards are the preferred form of incentives, emphasizing the importance of customizable reward systems in chore management solutions.

Desired Features

Respondents expressed strong interest in features that could simplify and automate the chore management process. The most requested features included:

- Customizable reward systems (81.3%) – Users want flexibility in how they incentivize task completion.
- Automated task assignments based on a child's age (68.8%) – A structured approach to delegating chores.
- Integration with calendars and reminders (43.8%) – To assist with scheduling and accountability.
- Gamification (43.8%) – Using badges and points to encourage engagement.
- Progress tracking and reports (31.3%) – Providing insights into completed and pending tasks.
- Group task assignment (18.8%) – Useful for teachers and households with multiple children.

Method 2: Interview

We conducted interviews with six participants, including parents, teachers, and couples, to explore their current chore management practices, challenges, reward systems, and desired features in a digital solution. Additionally, we used an affinity diagram to identify common themes and patterns across participants. These insights provide valuable guidance for designing an intuitive and efficient chore management tool.

Current Chore

Interview responses revealed that most participants do not use digital tools for chore management, instead relying on verbal reminders, whiteboards, or physical lists (such as Google Sheets or printed task lists). Chores are typically assigned based on preference or necessity, with couples often dividing responsibilities informally. Teachers managing classroom chores tend to use verbal instructions alongside a reward system to encourage participation. Some participants mentioned that time constraints make it difficult to maintain consistency in task management.

Challenges

Interview findings showed that most participants do not use digital tools for organizing chores. Instead, they rely on verbal reminders, whiteboards, or physical lists (such as Google Sheets or printed task lists). Chores are typically assigned based on preference or necessity, with couples informally dividing responsibilities. Teachers managing classroom tasks often use verbal instructions along with a reward system to encourage participation. Some participants mentioned that time constraints make it difficult to maintain consistency in task management.

Rewards

Reward systems varied significantly among participants. Parents and teachers frequently used gamification techniques, such as star-based rewards, leader privileges, and playtime incentives, to motivate younger children. Others relied on screen time or outings as a reward, while some preferred not to use any incentives, instead emphasizing responsibility. One participant mentioned occasionally using mild negative reinforcement, such as delaying dessert until tasks were completed. These findings suggest that a highly customizable reward system would be the most effective way to meet diverse user needs.

Desired Features

Participants expressed strong preferences for simplicity, ease of use, and automation in a chore management app. Commonly requested features included:

- A shared calendar and task reminders to improve accountability.
- Gamified elements like badges, points, and leaderboards to encourage engagement.
- A clear progress tracking system that provides visibility into completed and pending tasks.
- Voice-to-text capabilities to make task assignment more intuitive.
- Affordability, as some participants were concerned about potential subscription fees or ads. These results highlight the importance of a user-friendly, engaging, and cost-effective digital solution.

The full set of user personas derived from these insights can be found in Appendix: [User Personas](#).

Method 3: Usability Testing

Round 1: Usability Testing (Class Activity)

In the first round of usability testing, eight participants evaluated the system, focusing on core tasks such as assigning tasks, adding/deleting child profiles, and creating rewards. The overall usability scores varied depending on the task. (see Appendix: [Usability Testing Round 1 Result](#))

- Task 1: Assign Task – Average usability score (ASQ): 5.23
- Task 2: Add/Delete Child – ASQ: 4.10
- Task 3: Create a Custom Reward – ASQ: 5.90
- Task 4: Send a Reminder for Overdue Task – ASQ: 5.76

Key Findings & Issues:

The biggest issue identified was related to adding and deleting a child profile. Users struggled to locate the "Kid's Profile" section, because it was confused with the general Profile section. Also, the delete and edit icons were unclear, with many participants expecting to use the trash icon to delete a child profile. However, deletion required selecting the Edit option first, leading to confusion. Other tasks such as assigning tasks and sending reminders received moderate usability scores but did not present major usability concerns.

Round 2: Target User Usability Testing

Following the feedback from Round 1, design improvements were made to address the profile navigation issues and clarify the delete/edit function. The second round of usability testing was conducted with four target users (parents, teachers, and guardians) to validate the improvements and identify any new challenges. (see Appendix: [Usability Testing Round 2 Result](#))

- Task 1: Assign Task – ASQ: 3.70
- Task 2: Add Child – ASQ: 5.76
- Task 3: Delete Child – ASQ: 6.30
- Task 4: Create a Custom Reward – ASQ: 6.83
- Task 5: Send a Reminder for Overdue Task – ASQ: 6.70

Key Findings & Improvements:

Significant improvements were observed in the "Add/Delete Child" tasks after refining the profile section and making deletion more intuitive. ASQ scores for these tasks increased from 4.10 (Round 1) to 5.76 (Add Child) and 6.30 (Delete Child) in Round 2 (see Appendix: [Heat Map](#)). However, Assigning Tasks became the most challenging task, with an ASQ score decreasing to 3.70. Users reported difficulty finding where to assign a task, as they did not realize that clicking the "+" button was required. Some participants suggested incorporating a brief tutorial or onboarding guide to assist first-time users. Other tasks, such as creating custom rewards and sending reminders, showed overall positive improvements, with higher ASQ scores indicating greater ease of use.

Discussion

The findings from the project demonstrate that we successfully addressed its main objectives of improving household chore management. The usability testing, survey data, and interviews provided valuable insights into user behavior, pain points, and feature expectations.

Evaluation of Project Goals

The usability testing revealed that while most features were effective, the task assignment process needed refinement, as users struggled to locate the "+" button. Profile navigation issues identified in Round 1 were significantly improved in Round 2, as reflected in higher ASQ scores for adding and deleting children. Surveys and interviews confirmed that a customizable reward system was preferred over financial incentives, with parents and teachers valuing flexibility in assigning points, screen time, and privileges. Expanding the participant pool to include

teachers, couples, and guardians provided a broader perspective, ensuring that KidsConnect effectively addressed diverse household and classroom chore management needs.

Implications of Design

- Gamification Enhances Engagement: Positive feedback on gamified elements highlights the importance of engagement-driven task management. Features like badges, leaderboards, and points help motivate children to complete tasks.
- Task Assignment Clarity Needs Improvement: Users struggled with task assignment, suggesting the need for UI enhancements or onboarding features such as a tutorial or guided walkthrough for first-time users.
- Affordability Concerns: Participants expressed concerns about subscription fees or ads, emphasizing that affordability is key. A freemium model offering essential features for free with premium customization options could improve adoption.

What Could Have Done Differently

If we had more time, we could have conducted usability testing with children to gain insights into how they interact with the system, ensuring that features like gamification and task assignment are engaging and intuitive for them. Additionally, we could have tested with more participants to strengthen our findings, gathering a broader range of perspectives and validating usability improvements across different demographics. Expanding the number of interviews with parents, teachers, and caregivers would have provided deeper qualitative insights, allowing us to refine features based on a more diverse set of user needs.

Opportunities for Future Work

- Improved Onboarding Experience: Adding a tutorial or interactive guide would help first-time users understand key features, especially the task assignment process.
- AI-Powered: Implementing AI task recommendations could assist parents and teachers in assigning chores more efficiently by analyzing user habits and preferences.
- Real-Time Collaboration: Introducing multi-user synchronization would allow multiple family members to update and track tasks in real-time, improving engagement and accountability.

Conclusion

In conclusion, KidsConnect's main goal is to provide parents, teachers, and caregivers with an easy way to assign and track tasks through a gamified and customizable task reward system with a user-friendly interface. The research conducted, including surveys, interviews, and usability testing, validated many of the key findings from the literature review and competitive analysis. The improvements made in the second round of testing showed significant results, advancing the project toward final refinements and a more user-centered approach.

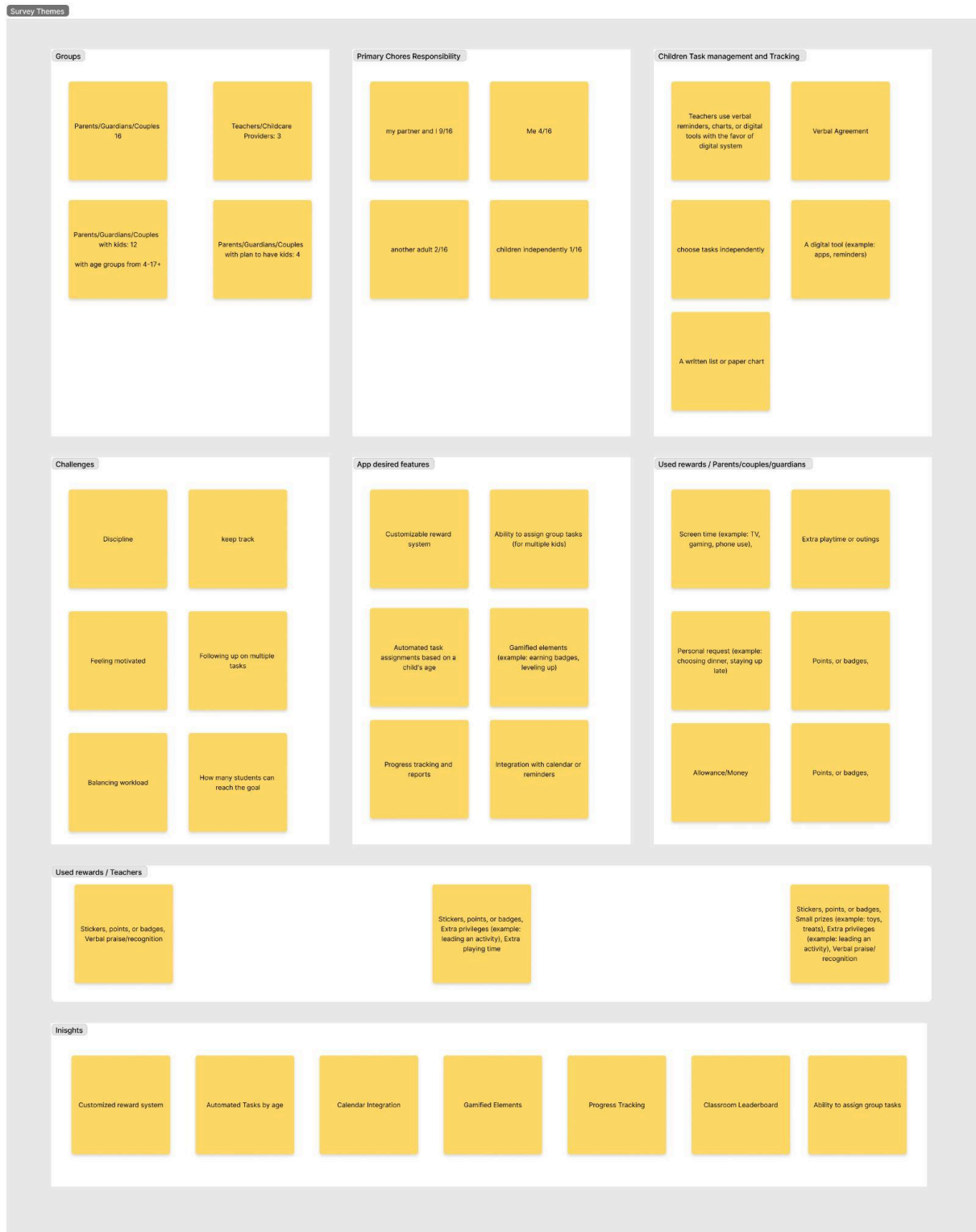
Future work could focus on adding a tutorial to the onboarding process, incorporating AI-powered task recommendations based on age, and expanding usability testing to include children and cover a larger number of participants. This would help validate and enhance the reward system, allowing KidsConnect to reach its full potential as an essential tool for fostering responsibility, collaboration, and motivation in children's daily routines.

References

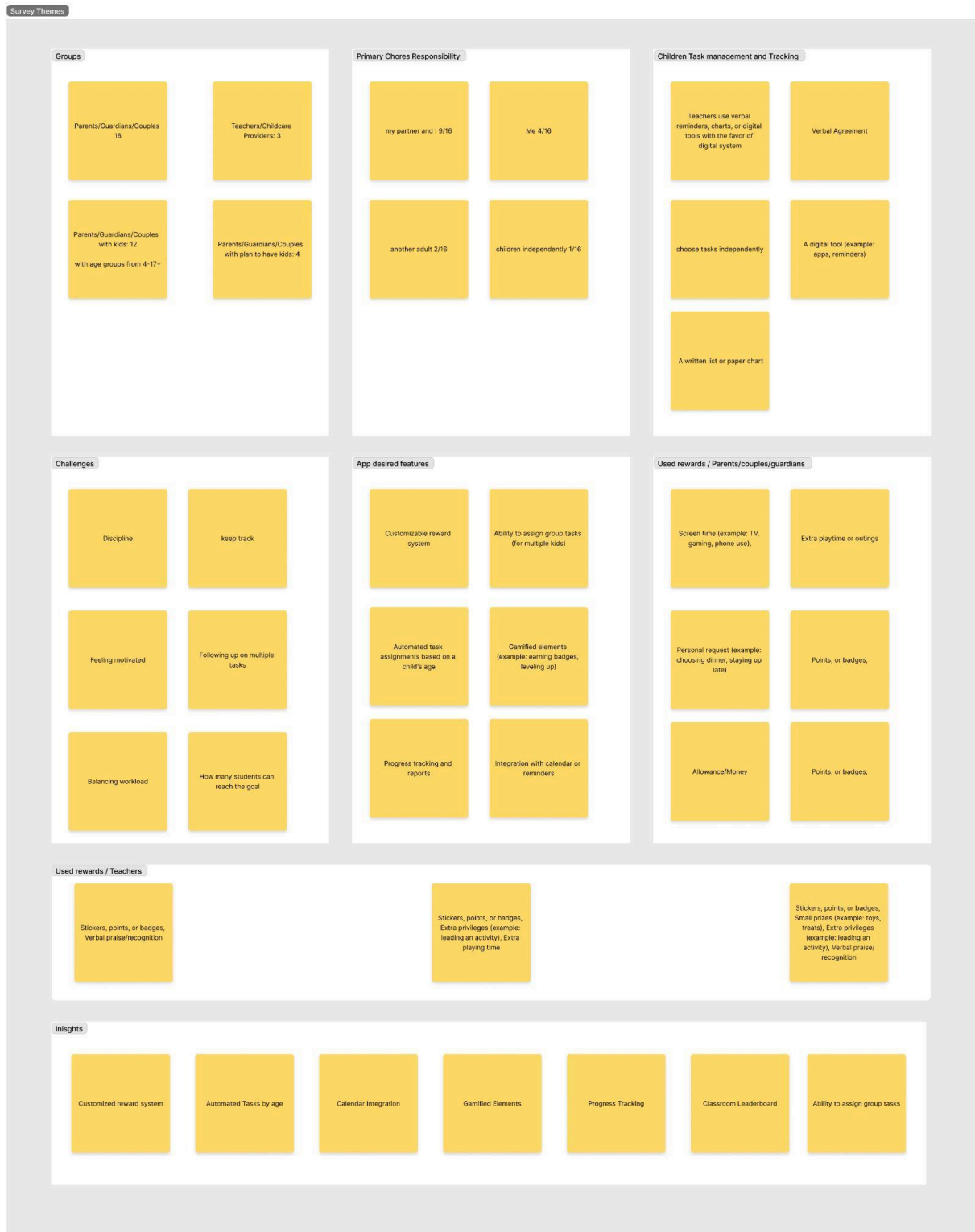
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Appendix 1: Supporting Material

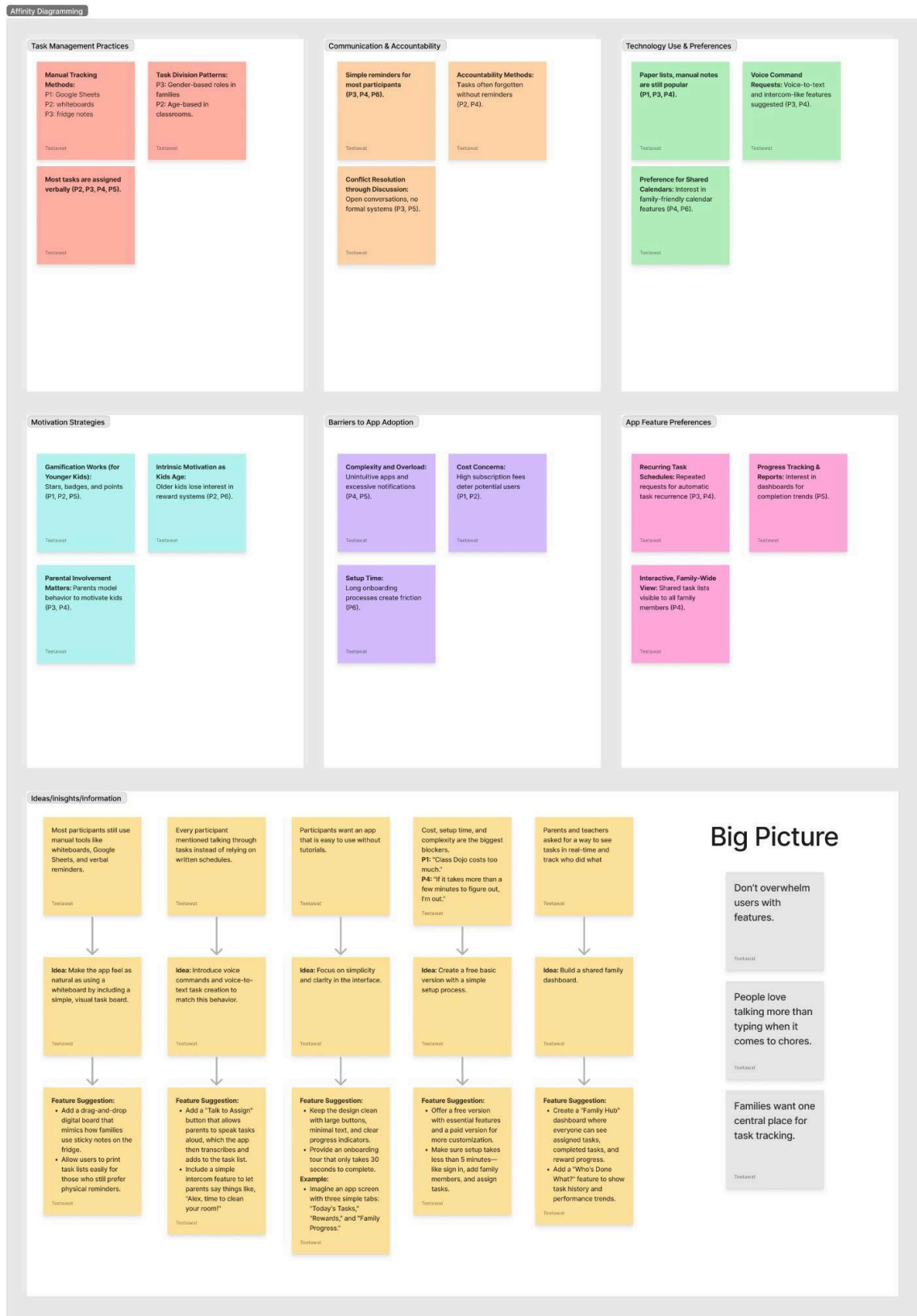
Survey Theme



Interview Coding



Affinity Diagram



User Personas

Basic Information



Emily Carter

"The Busy Parent"

- Age: 35
- Occupation: Marketing Manager
- Family: Married, two kids (ages 6 & 10)
- Location: Chicago, IL
- Tech Comfort: Moderate

Bio

Emily is a working mom juggling her career and household responsibilities. She struggles to keep her kids accountable for chores and often resorts to verbal reminders or sticky notes on the fridge. She has tried using shared calendars but finds them too manual to maintain. She needs a simple, efficient system to track chores and encourage her kids to participate—without spending too much time setting it up.

More about Emily Carter

Pain Points

- Kids forget chores without constant reminders.
- No easy way to track completed vs. pending tasks.
- Reward systems are sticky charts that work at first but lose effectiveness.
- More voice or quick-entry methods rather than typing everything.

Goals & Needs

- A quick and easy way to assign and track chores.
- A shared family dashboard where kids can see their tasks.
- Non-financial rewards like extra screen time or special privileges.
- A way to see who's doing the household work to ensure fairness.

Basic Information



James Anderson

"The Organized Teacher"

- Age: 42
- Occupation: 4th Grade Teacher
- Family: Married, one daughter (4 years old)
- Location: Chicago, IL
- Tech Comfort: High

Bio

James teaches a classroom of 20+ students and often assigns classroom responsibilities (cleaning desks, organizing supplies, etc.). However, tracking who completed what is a hassle. He currently uses a paper-based checklist and sometimes a whiteboard system, but students often forget their assigned tasks. He is open to using an app, but it must be simple and adaptable for different classroom settings.

More about James Anderson

Pain Points

- Kids forget classroom chores without direct supervision.
- No digital record to track task completion over time.
- Different teachers assign tasks differently, leading to inconsistencies.
- Students lose interest in basic reward systems.

Goals & Needs

- A task tracking system that students can check independently.
- A class-wide visibility dashboard so everyone sees assigned tasks.
- A badge/reward system to keep students motivated.
- A multi-teacher system where different teachers can assign and monitor tasks.

Usability Testing Round 1 Result

Assign a Task to a Child Prototype Test - Screen-based Imagine you want to assign a new task to two children, please make sure to fill all the fields.			48.1s Avg. duration	11 Responses
 90.9% Success rate	 9.1% Drop-off	 39.8% Misclick rate		
Add/Remove Children Profiles Prototype Test - Screen-based You need to remove a child profile from the application and then add a new child profile, please go ahead and start the task.			51.4s Avg. duration	10 Responses
 100.0% Success rate	 0.0% Drop-off	 55.5% Misclick rate		
Create a Custom Reward Prototype Test - Screen-based Imagine you want to create a custom reward that suits the children's needs, please make sure to fill all the fields.			20.2s Avg. duration	10 Responses
 100.0% Success rate	 0.0% Drop-off	 35.4% Misclick rate		
Send an Alert for Overdue Tasks Prototype Test - Screen-based You notice that one of your children has not completed their assigned task, and you want to send them a reminder.			18.0s Avg. duration	10 Responses
 100.0% Success rate	 0.0% Drop-off	 34.2% Misclick rate		

ASQ questionnaire Round 1

Task 1: Assign a Task to a Child (The average score is 5.23)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree			10 Responses	5.3 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree			10 Responses	5.4 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree			10 Responses	5 Average

Task 2: Add/Remove Children Profile (The average score is 4.10)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	3.8 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	4.5 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	4 Average

Task 3: Create a Custom Reward (The average score is 5.90)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	5.8 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	5.9 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	6 Average

Task 4: Send an Alert for Overdue Tasks (The average score is 5.76)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	5.7 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	5.9 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	10 Responses	5.7 Average

Usability Testing Round 2 Result

Assign Task to Children Prototype Test · Screen-based You are a caregiver responsible for managing daily tasks for children. Your goal is to quickly assign the "Make Bed" task to Maya and Jack using the "Recently Assigned Task" feature.	116.9s Avg. duration	4 Responses
100.0% Success rate	0.0% Drop-off	49.1% Misclick rate
Add Child Profile Prototype Test · Screen-based You are a caregiver setting up task management for children and need to add a new child profile. Navigate to the Add Child section, enter the required details, and complete the process.	516s Avg. duration	4 Responses
100.0% Success rate	0.0% Drop-off	40.6% Misclick rate
Delete Child Profile Prototype Test · Screen-based You are a teacher managing children's profiles and need to delete Mishelle from the system. Navigate to the child management section, locate Mishelle's profile, and delete it.	9.5s Avg. duration	4 Responses
100.0% Success rate	0.0% Drop-off	29.4% Misclick rate
Add a Custom Reward Prototype Test · Screen-based You are a parent managing rewards for your children and need to add a new custom reward. Navigate to the rewards section, select the option to add a custom reward, enter the details, and create it.	22.9s Avg. duration	4 Responses
100.0% Success rate	0.0% Drop-off	37.5% Misclick rate

ASQ questionnaire Round 2

Task 1: Assign a Task to a Child (The average score is 3.70)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	3.8 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	4.3 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	3 Average

Task 2: Add Child Profile (The average score is 5.76)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	5.8 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	6.5 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	5 Average

Task 3: Delete Child Profile (The average score is 6.33)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	6.5 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	6.5 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	6 Average

Task 4: Add a Custom Reward (The average score is 6.83)

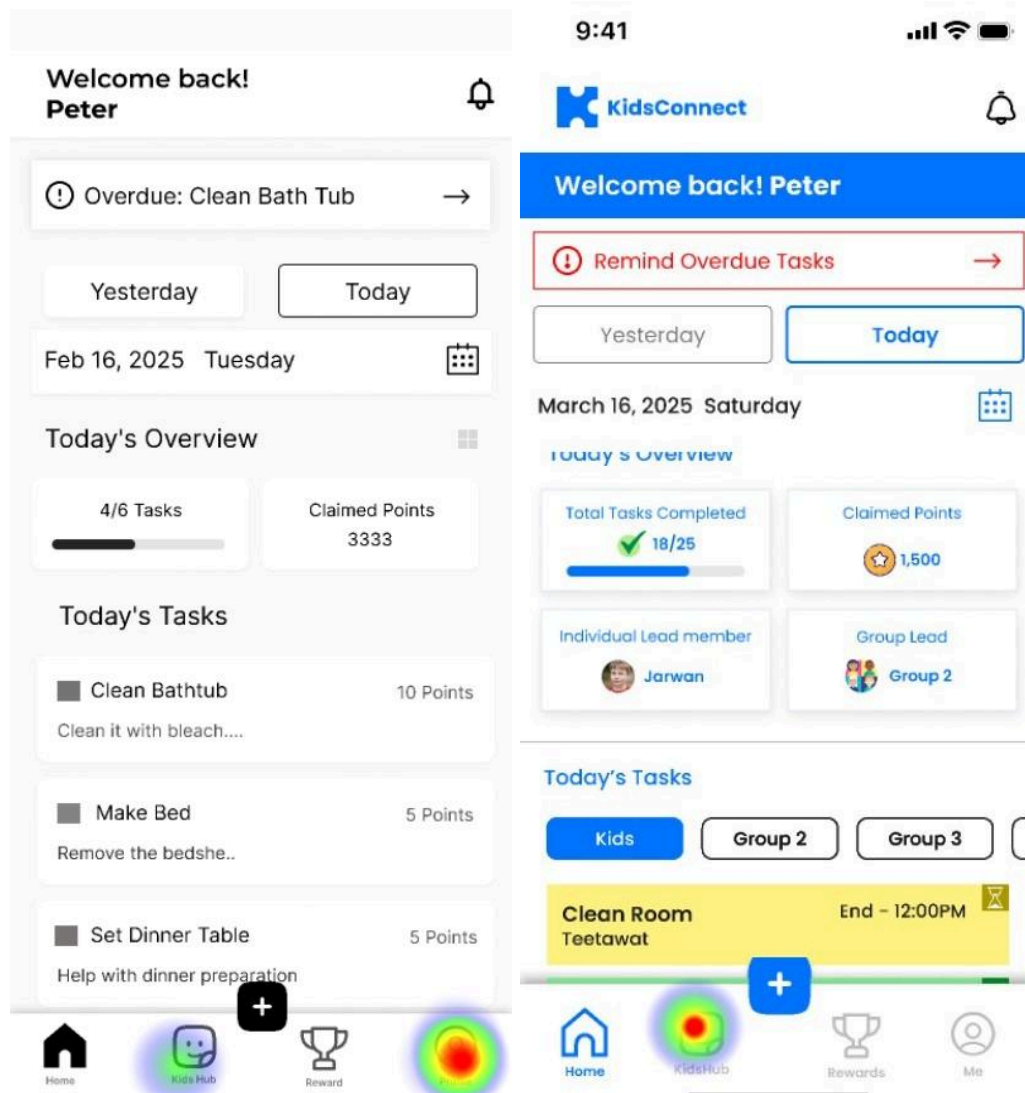
Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	6.5 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	7 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses	7 Average

Task 5: Send a Reminder to Overdue Task (The average score is 6.70)

Overall, I am satisfied with the ease of completing the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses 6.5 Average
Overall, I am satisfied with the amount of time it took to complete the task in this scenario. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses 6.8 Average
Overall, I am satisfied with the support information (on-screen guidance, error messages, documentation) when completing the task. Opinion Scale 1 = Strongly Disagree 2 = Disagree 3 = Somewhat Disagree 4 = Neutral 5 = Somewhat Agree 6 = Agree 7 = Strongly Agree	4 Responses 6.8 Average

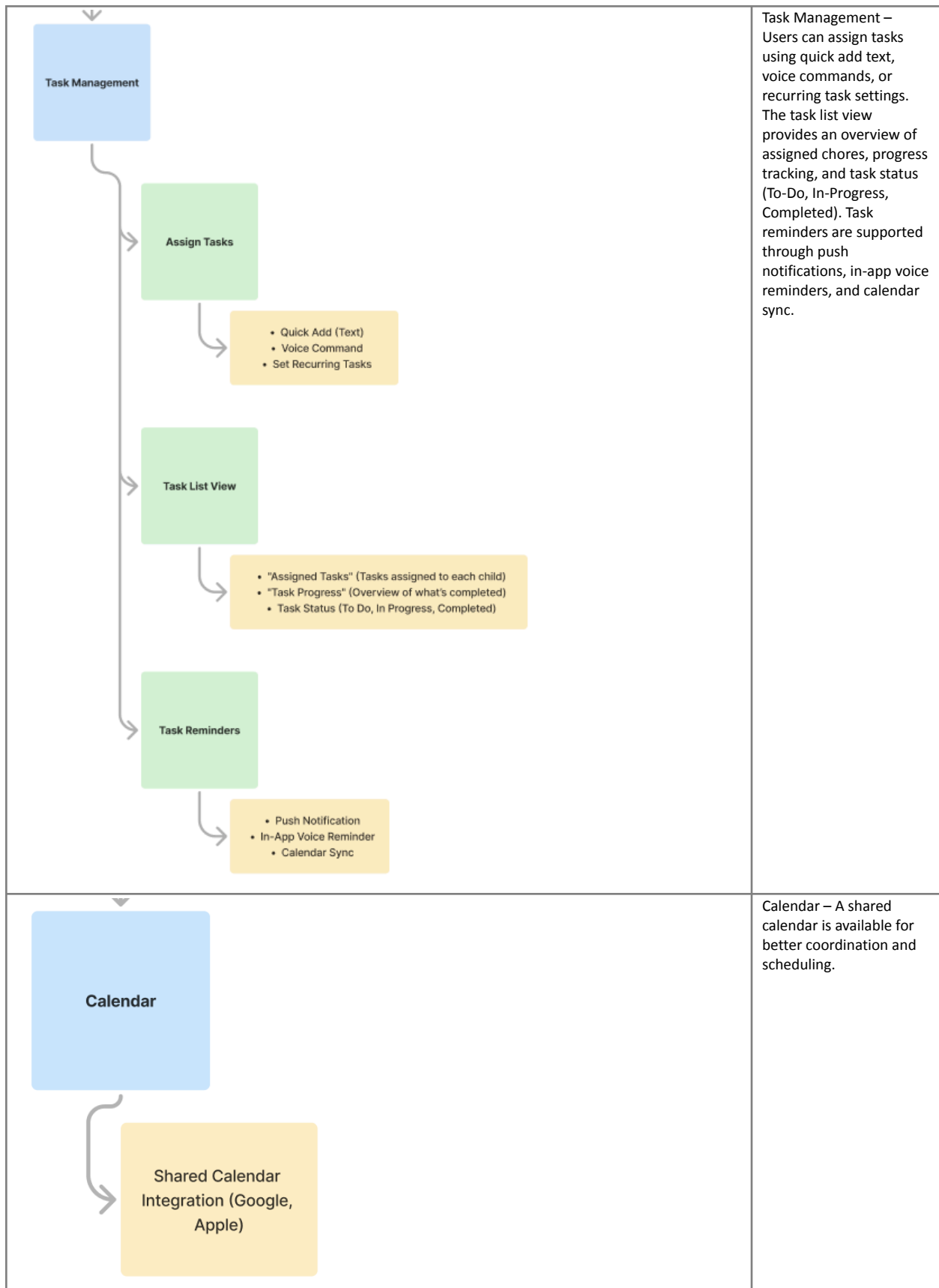
Heat Map for adding/deleting child profile Scenario

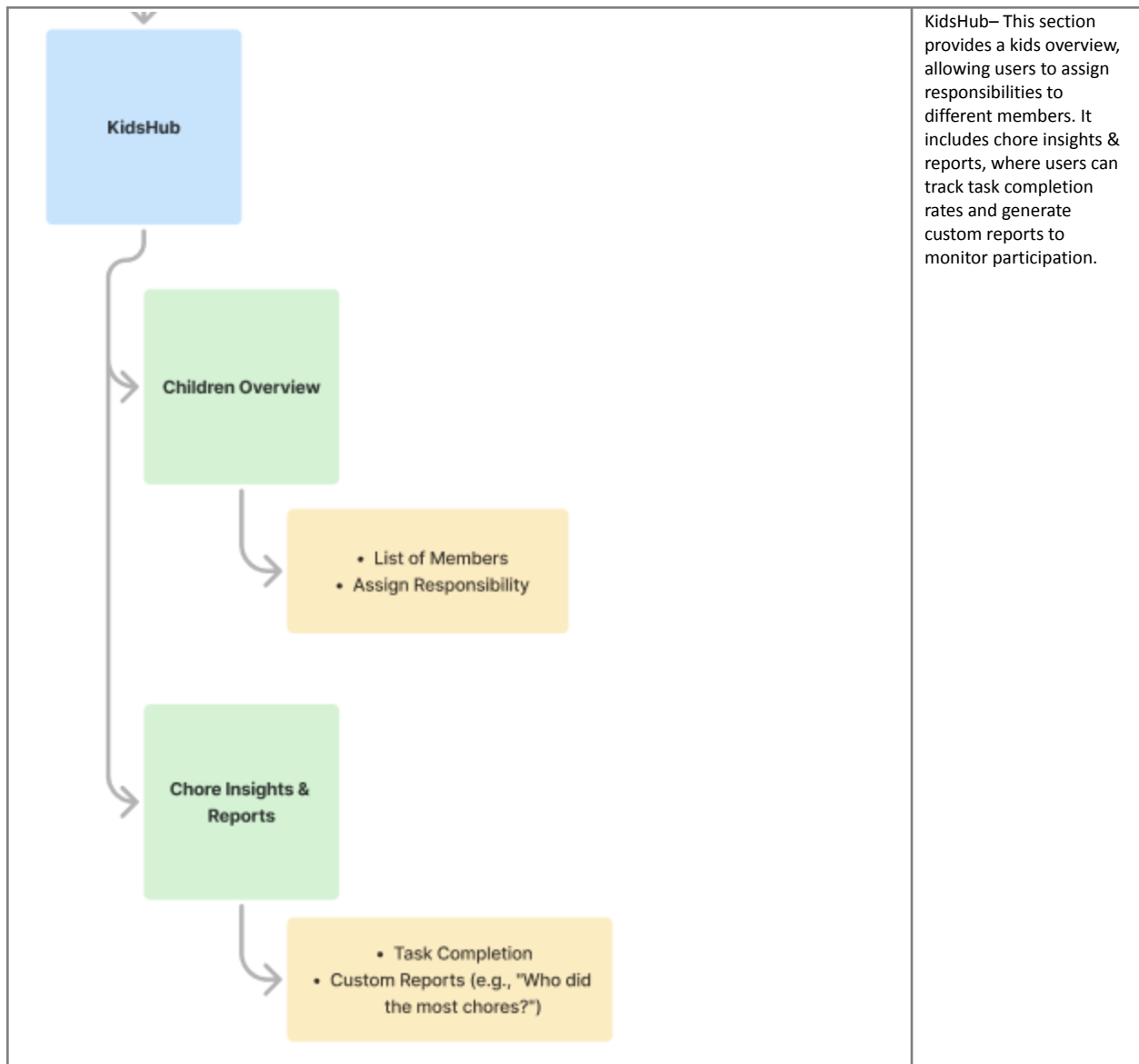
On the left is the old heat map in the first user test. on the right is the second heat map in the second user test.

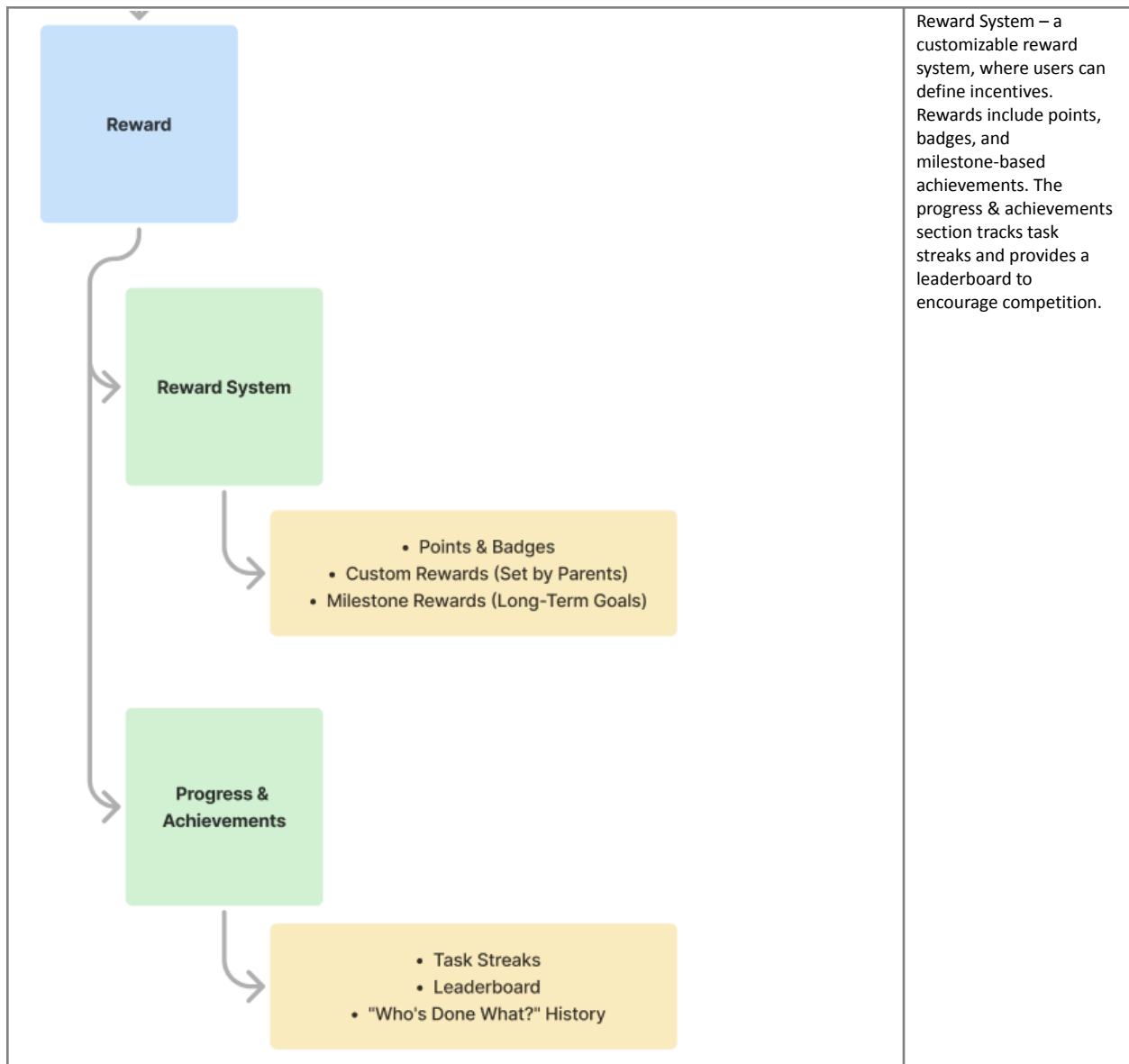


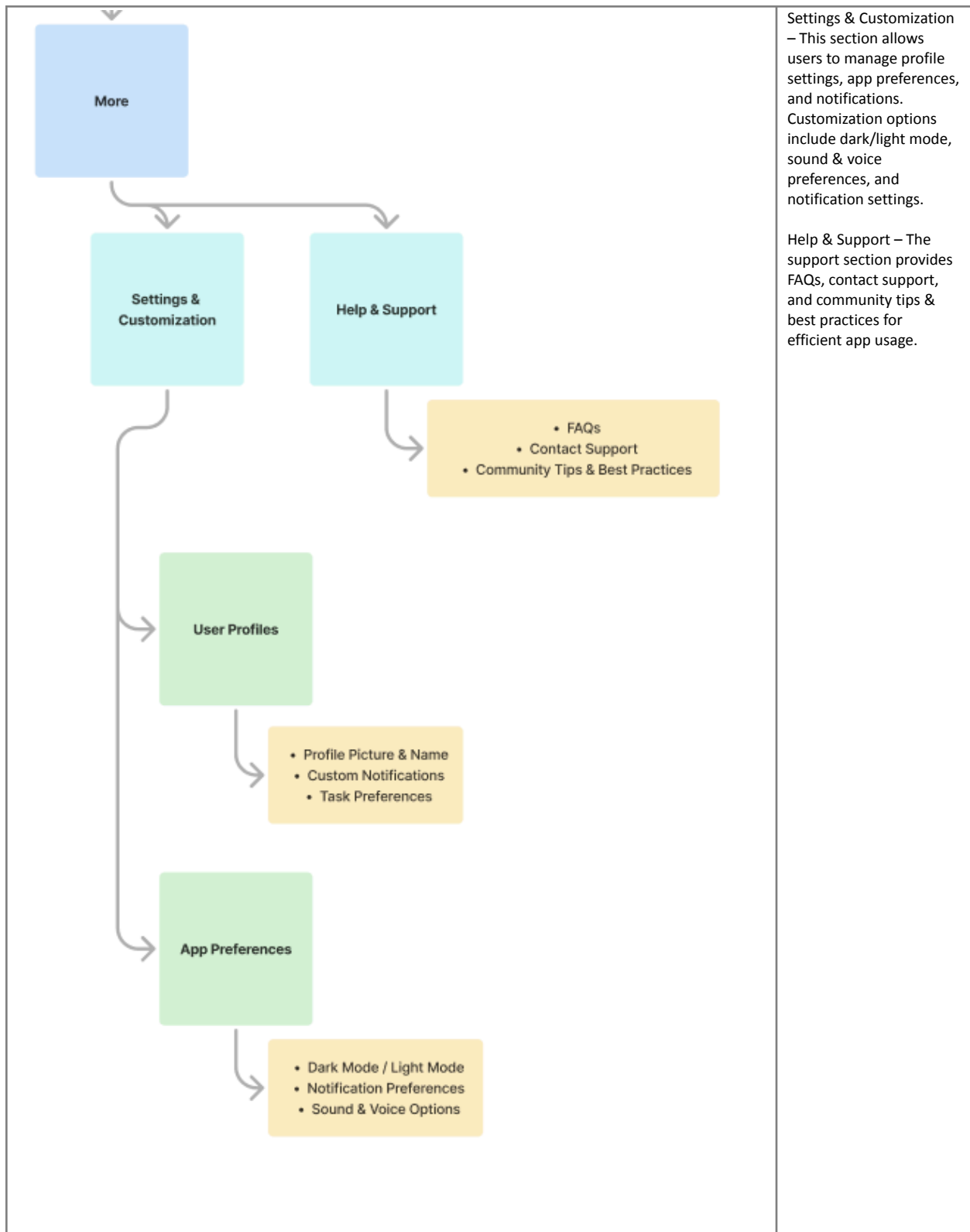
Site Map or Flow Diagram

Site Map	Annotations
Link: https://www.figma.com/board/V5WXq23LLVvk8urzfuCoA4o/HCI-CapStone-Coding%2Faffinity%2Fsurvey?node-id=99-140&t=QavVPaNCkaI05IR3-1	The sitemap is structured around a home dashboard, which serves as the central hub for navigation. The site map outlines the key sections of the application, each catering to specific functionalities for task management, family collaboration, and rewards.









Interactive Prototype

Prototype Links/Instructions

Prototype link: [Prototype Link](#)

Prototype instructions:

Home Screen

The prototype illustrates the Home Screen layout with the following components:

- Header:** Status bar (9:41, signal, battery), KidsConnect logo, and a notification bell.
- Welcome:** "Welcome back! Peter" with a blue bar and a small '1' annotation.
- Reminder:** "Remind Overdue Tasks" with a red arrow and a small '1' annotation.
- Calendar:** "Yesterday" and "Today" buttons, with "Today" selected. The date is "March 16, 2025 Saturday" with a small '2' annotation.
- Today's Overview:** Section with four cards: "Total Tasks Completed" (18/25), "Claimed Points" (1,500), "Individual Lead member" (Jarwan), and "Group Lead" (Group 2). A small '3' annotation is near the points card.
- Today's Tasks:** Section with tabs for "Kids", "Group 2", and "Group 3". A list of tasks follows: "Clean Room Teetawat" (End - 12:00PM), "Clean Bath Jarwan" (End - 10:00AM, checked), "Clean Room Sai" (End - 12:00PM, overdue), "Clean Bath Mishelle" (End - 10:00AM, checked), "Clean Bath Jarwan" (End - 10:00AM, checked), and "Clean Room Sai" (End - 12:00PM). A small '4' annotation is near the tabs.
- Bottom Bar:** Navigation icons for Home, KidsHub, Rewards, and Me, with a central "+" button. A small '5' annotation is near the "+" button.

1- A quick access to overdue tasks to send reminders, under is the calendar and the day to present home screen data either today or yesterday.

2- The calendar will expand and have the options to sync with phone calendar and export calendar for different uses.

3- Overview display of different status based on the date.

4- Tasks assigned today and their status. ability to choose different groups or kids to view what they have done or yet to be done.

5- Main Navigation includes kids hub, rewards, and me page. Also, the plus blue icon is for a quick access to create a new task.

Assign Task

← Assign Task

×

Recently Assigned Task

1

Make Bed
5 Points

Clean Bed
5 Points

Make
5 Po

Title

2

Enter your title here..

Description

Enter description...

Start Date

Start time

mm/dd/yyyy

HH:MM

End Date

End time

mm/dd/yyyy

HH:MM

Points

−

0 points

+

Repeat

☐

3

Select Assignees

>

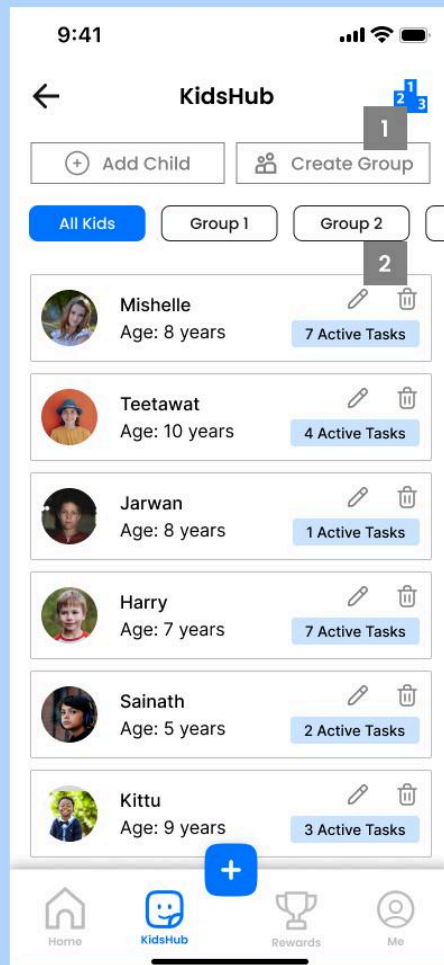
Assign Task

1- Recently assigned task feature for a quick task assigning based on task history.

2- Task information to be filled if a custom task is being created. if the recent task is chosen the fields will be filled automatically.

3- Select Assignees(children) that will be assigned the created task.

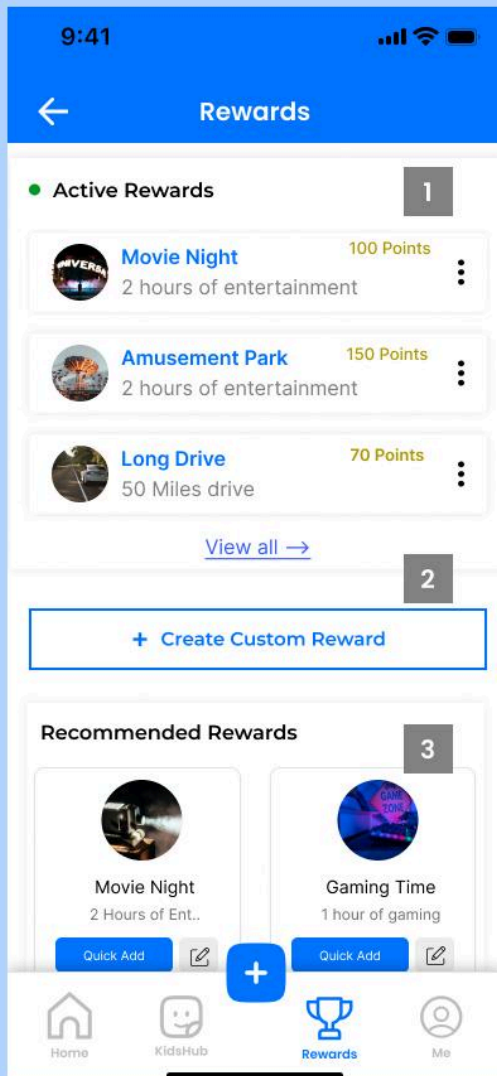
Kids Hub



1- Leader Board icon to see overall status of kids achievements. Add child and create group buttons.

2- Added children with name, age, number of active tasks. The ability to edit or delete a child profile beside each child.

Rewards



1- Active rewards, it will show what are the current active rewards with information about it.

2- Create custom reward button will pop up a screen to create a custom reward when filling the needed fields.

3- Recommended Rewards is to quickly add a new reward based on personal recommendation overtime.

User Interview Template/Script

Interview Protocol (Teacher)

Intro	Hello, my name is [Your Name], and my team is conducting interviews to understand how individuals manage shared responsibilities in their households or classrooms and what features would be useful in a chore management application. There are no right or wrong answers. This interview should take approximately 30 minutes. Please try to avoid distractions during this session The goal of this study is to identify the challenges individuals face in managing shared responsibilities and to design a tool that makes the process easier. Your responses will be anonymized and used only for research purposes. I will ask you a series of questions regarding how you manage shared responsibilities, the tools you currently use, and any challenges you experience. I may ask follow-up questions to clarify certain points. The interview will be recorded and transcribed for analysis. Before we continue, I need your informed consent. This form explains the purpose of the study, data confidentiality, and your rights as a participant. Participation is voluntary, and you can withdraw at any time. Provide the informed consent form.
Warm-up and build rapport	Please tell me a little about yourself. What age group or grade level do you teach? How do you currently manage classroom chores and responsibilities?
Deep Focus	Chore Management Can you walk me through how you assign classroom chores or responsibilities to students? How do you ensure that students complete their assigned tasks? What challenges do you face when it comes to keeping students accountable for their chores? Have you tried using any tools (apps, planners, whiteboards) to manage classroom tasks? If yes, what worked well and what didn't? Communication & Accountability How do you remind students about their tasks? Do you use any reward or consequence system to encourage students to complete their tasks? What methods have been most effective in ensuring classroom tasks get done? App Preferences If you could design the perfect app to help with managing classroom chores, what features would it have? What are some deal-breakers that would make you not want to use a classroom chore management app? How comfortable are you with using mobile or desktop applications for organizing tasks?
Restrospective	If you had access to a task management tool, how would it change your current routine? What's the biggest frustration you experience in managing classroom responsibilities? What are the most important features for an app like this to have?
Demographic	What is your age? What subjects do you teach? On a scale of 1-5, how comfortable are you with using technology to manage tasks?
Wrap-Up	Thank you, [participant's name], for participating in our study! Your feedback is incredibly valuable. Before we wrap up, do you have any final thoughts or questions for me?

Interview Protocol (Couple)

Intro	Hello, my name is [Your Name], and my team is conducting interviews to understand how individuals manage shared responsibilities in their households or classrooms and what features would be useful in a chore management application. There are no right or wrong answers. This interview should take approximately 30 minutes. Please try to avoid distractions during this session The goal of this study is to identify the challenges individuals face in managing shared responsibilities and to design a tool that makes the process easier. Your responses will be anonymized and used only for research purposes. I will ask you a series of questions regarding how you manage shared responsibilities, the tools you currently use, and any challenges you experience. I may ask follow-up questions to clarify certain points. The interview will be recorded and transcribed for analysis. Before we continue, I need your informed consent. This form explains the purpose of the study, data confidentiality, and your rights as a participant. Participation is voluntary, and you can withdraw at any time. Provide the informed consent form.
Warm-up and build rapport	Please tell me a little about yourself. How do you currently keep track of household chores? (e.g., lists, verbal reminders, apps, etc.) Are there any specific strategies or routines you follow to ensure chores get completed?
Deep Focus	Chore Management Can you walk me through a typical week in managing household chores? How do you and your partner divide chores? Is it based on preference, time availability, or something else? What challenges do you face when it comes to getting chores completed on time? Have you tried using any tools (apps, planners, whiteboards) to manage household tasks? If yes, what worked well and what didn't? Have you started thinking about how chore management will change when you have children? Do you plan to assign specific chores to your future children? If so, how do you envision that working Communication & Accountability How do you and your partner remind each other about household tasks? Have you ever had disagreements about chores? If so, how do you resolve them? Do you have any incentives or systems in place to encourage task completion? How do you anticipate managing accountability when you have children? App Preferences If you could design the perfect app to help with chores, what features would it have? What are some deal-breakers that would make you not want to use a chore management app? How comfortable are you with using mobile applications for organizing tasks?
Retrospective	If you had access to a task management tool, how would it change your current routine? What's the biggest frustration you experience in managing household chores? What are the most important features for an app like this to have? How do you expect your household task management needs to change when you have children?
Demographic	What is your age? What is your occupation? On a scale of 1-5, how comfortable are you with using technology to manage tasks?
Wrap-Up	Thank you, [participant's name], for participating in our study! Your feedback is incredibly valuable. Before we wrap up, do you have any final thoughts or questions for me?

Interview Protocol (Parent)

Intro	Hello, my name is [Your Name], and my team is conducting interviews to understand how individuals manage shared responsibilities in their households or classrooms and what features would be useful in a chore management application. There are no right or wrong answers. This interview should take approximately 30 minutes. Please try to avoid distractions during this session. The goal of this study is to identify the challenges individuals face in managing shared responsibilities and to design a tool that makes the process easier. Your responses will be anonymized and used only for research purposes. I will ask you a series of questions regarding how you manage shared responsibilities, the tools you currently use, and any challenges you experience. I may ask follow-up questions to clarify certain points. The interview will be recorded and transcribed for analysis. Before we continue, I need your informed consent. This form explains the purpose of the study, data confidentiality, and your rights as a participant. Participation is voluntary, and you can withdraw at any time. Provide the informed consent form.
Warm-up and build rapport	Please tell me a little about yourself. How do you currently keep track of household chores? (e.g., lists, verbal reminders, apps, etc.) Are there any specific strategies or routines you follow to ensure chores get completed?
Deep Focus	Chore Management Can you walk me through a typical day in managing household chores? How do you assign chores to your children or other household members? What challenges do you face when it comes to getting chores completed on time? Have you tried using any tools (apps, planners, whiteboards) to manage household tasks? If yes, what worked well and what didn't? Communication & Accountability How do you remind your children about their chores? Do you have any consequences or rewards for completing chores? What methods have been most effective in ensuring chores get done? App Preferences If you could design the perfect app to help with chores, what features would it have? What are some deal-breakers that would make you not want to use a chore management app? How comfortable are you with using mobile applications for organizing tasks?
Retrospective	If you had access to a task management tool, how would it change your current routine? What's the biggest frustration you experience in managing household chores? What are the most important features for an app like this to have?
Demographic	What is your age? How many children do you have, and what are their ages? What is your occupation? On a scale of 1-5, how comfortable are you with using technology to manage tasks?
Wrap-Up	Thank you, [participant's name], for participating in our study! Your feedback is incredibly valuable. Before we wrap up, do you have any final thoughts or questions for me?